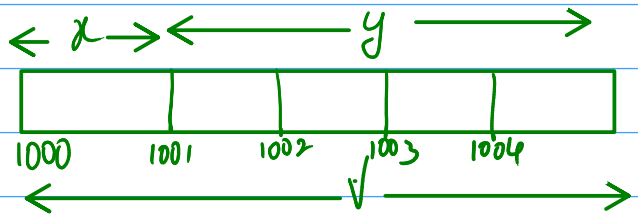
11/03

```
struct St
{
        char x;
        int y;
}v;
```

expected size 5 bytes

what we get is 8 bytes



What is CPU machine cycle?

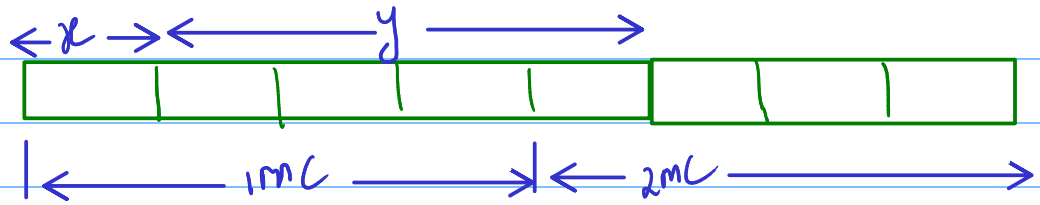
what is a word size?

word size may be 2bytes or 4bytes or 8 bytes based on compiler.

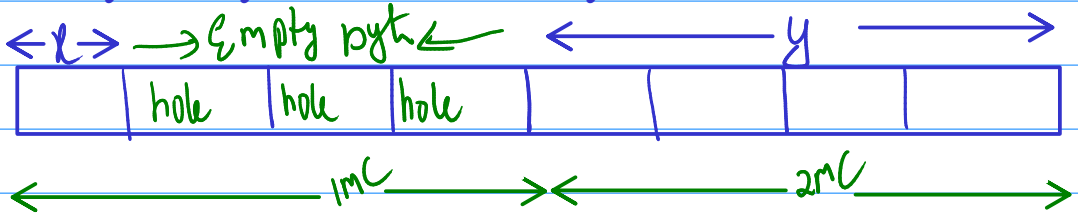
on 16bit compiler wordsize is 2bytes
on 32bit compiler wordsize is 4bytes
on 64 bit compiler wordsize is 8bytes

Lets us explore the example based on 32bit compiler

In 32bit compiler , one cpu machine cycle(1cpcmc) fetches 4bytes



to fetch the y value we need 2 cpu machiune cycles. which increases the work load of cpu, to make easy , compiler pushes the first 3 bytes of y to next machine cycle.



The Process of adding empty bytes between the members of structure of different datatypes is called structure padding.

Empty bytes are called as holes.

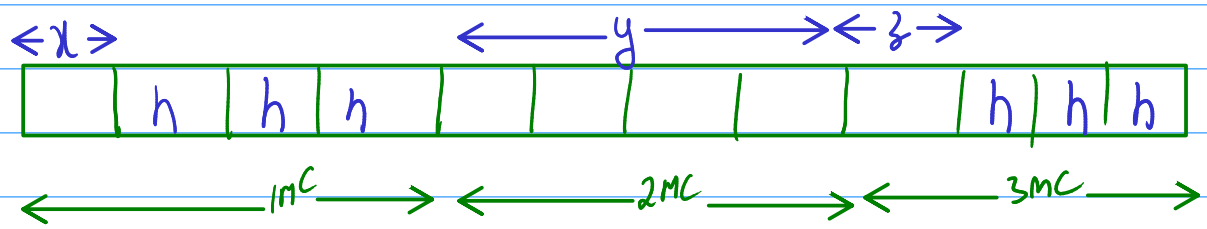
structure padding can be multiples of 2bytes or 4bytes or 8 bytes based on biggest member

Note : turbo C does not support structure padding.

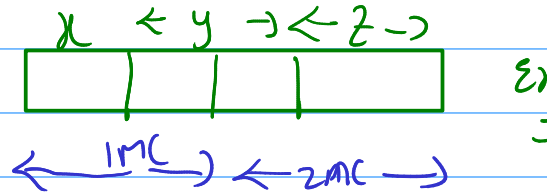
structure padding is automatically done by the compiler itself.

```
struct St
{
        char x;
        int y;
        char z;
};
```

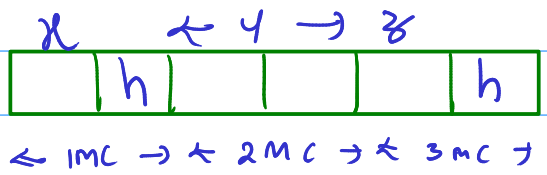
12bytes




```
struct St
{
        char x;
        short int y;
        char z;
};
```



Expected

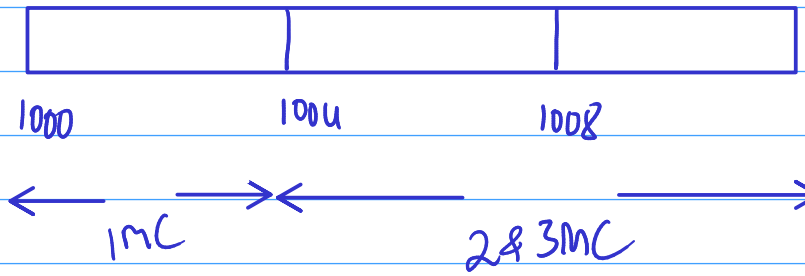


6bytes

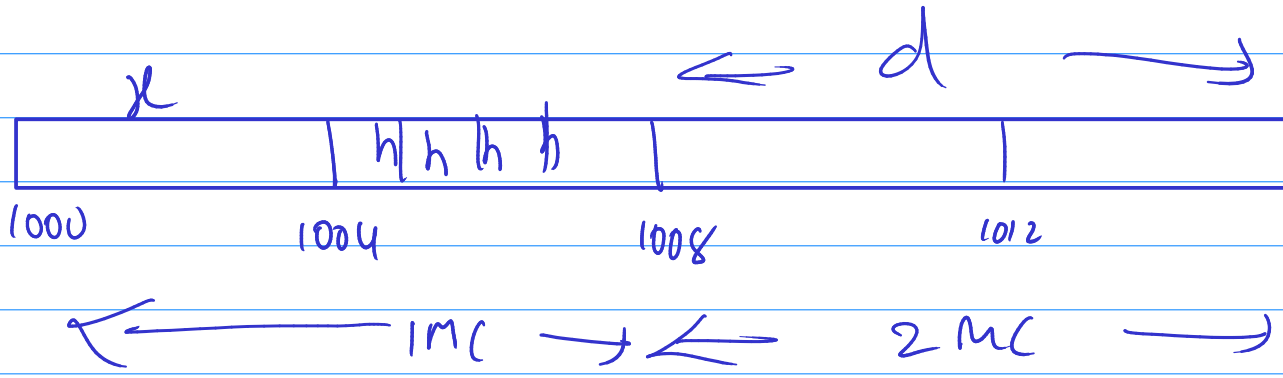
```
struct St
{
        int x;
        double d;
};
```

on 32 bit compiler

12 bytes



on 64 bit compiler



How to avoid padding?

1. declaring the members of structure in a proper alignment.
2. using structure packing
    a. #pragma pack
    b. attribute pack

```
struct St
{       char x;
        int y;
        char z;
};
```

12 byte

samething with proper alignment

```
struct St
{       char x;
        char z;
        int y;
};
```

8 byte